6INTELSY - Intelligent Systems

Final Project Speci0ications

2nd Semester, School Year 2025-2026

1) Overview

In **teams of 4**, design, implement, document, and **defend** a working AI system on GitHub. Your system must include:

1. A **core deep learning model** (vision, text, audio, time-series, or tabular).
2. A **CNN component** (may be the core model, or a supporting experiment such as text-CNN, TCN, or spectrogram CNN).
3. An **NLP component** (can be core or auxiliary—e.g., text classifier, NER, retrieval, summarizer, or metadata NLP).
4. A **small RL component** (e.g., contextual bandit, Q-learning, threshold-tuning agent, or simple environment policy).
5. A robust **ML pipeline**: dataset selection/governance, preprocessing, training, evaluation, ablations, error/slice analysis.
6. A concise **Ethics & Policy** analysis with concrete mitigations and a **Model Card**.

Platform: GitHub repository is **required** for submission, versioning, issues, and project board.

2) Team & Roles (maximum of 4 students)

- **Project Lead / Integration** – scope, timeline, integration, release tagging, defense coordination.
- **Data & Ethics Lead** – dataset sourcing, licensing/consent, preprocessing, bias checks, ethics statement.
- **Modeling Lead** – model design (CNN/NLP), training pipeline, ablations, error analysis.
- **Evaluation & MLOps Lead** – metrics, reproducibility, environment files, automation (scripts/Makefile/CI).

Everyone contributes to code, documentation, and the defense.

3) Project Model

This list **Task**, **MVP**, **Metrics**, **Stretch**, and **Ethics focus**.

1. **Personalized Recommendation with Contextual Bandits (NLP + RL) – Recommender Sys**
    - **Task:** Rank items/news using text embeddings; adapt via contextual bandit.
    - **MVP:** Embedding-based ranker + ε-greedy/LinUCB in offline simulation.
    - **Metrics:** nDCG/Hit@K (offline) + cumulative reward in sim.
    - **Stretch:** Diversity and exposure fairness constraints.
    - **Ethics:** Filter bubbles; privacy; consent for behavioral data.

4) Cross-Cutting Requirements (What must be in every project)

4.1 Data, Preparation & Training

- **Dataset governance:** source, license, consent/PII, representativeness; justify choice.
- **Splits:** train/val/test with **no leakage**; stratified/temporal as appropriate.
- **Baselines:** at least one simple **non-DL** baseline and one **DL** baseline.
- **Training:** fixed seeds, logged hyperparameters, early stopping/schedules.
- **Ablations (≥2 within 3 weeks):** e.g., augmentation on/off, optimizer, LR, depth/width.
- **Error/slice analysis:** confusion breakdowns, failure cases, relevant subgroups.

4.2 CNN Requirement

- Include **one CNN experiment** (core or support): e.g., ResNet/MobileNet, text-CNN, TCN, or spectrogram CNN.

4.3 NLP Requirement

- Include **one NLP task**: text classification, NER, retrieval, summarization, or tokenizer-level analysis supporting the main pipeline.

4.4 RL Requirement

- Include **one RL component**: simple Q-learning/DQN, policy gradient, contextual bandit, or an agent for threshold/policy tuning. Provide environment spec, reward function, and learning curves.

4.5 Ethics & Policy

- **Ethical Impact Statement** (1–2 pages) with risks, mitigations, intended use/limitations, privacy & consent, fairness checks, and misuse considerations.
- **Model Card** summarizing data, usage, metrics, slices, and caveats.

4.6 Reproducibility & Engineering

- **One-command reproduce**: make repro or bash run.sh to train/evaluate and regenerate key results.
- **Environment file:** requirements.txt or environment.yml.
- **Runtime limit:** choose settings so full training/evaluation completes in **≤ 90 minutes** on Colab/Kaggle or a single mid-range GPU.

5) Deliverables & 3-Week Timeline

All work is submitted via GitHub (repo link). Use Issues and a Project board for tasks. Tag releases as **v0.1 (proposal)**, **v0.9 (RC)**, **v1.0 (final)**.

Week 1 — Proposal & Setup (Due end of Week 1)

- **Proposal (2–3 pages in docs/):**
  - Problem statement & scope; success metrics and constraints.
  - Related work (3–6 citations).
  - Dataset plan (source, license, consent, representativeness).
  - Planned CNN/NLP/RL components and MVP.
  - Ethics risk register (top 3 risks + mitigations).
  - Team roles & milestones.
- **GitHub repo initialized:** README, environment, data download script/template, project board.
- **Deliver:** GitHub Release **v0.1**, Proposal PDF, and short **2–3 min** screencast (optional but recommended).

Week 2 — Checkpoint (Due end of Week 2)

- Data acquired/cleaned; splits finalized; EDA notebook.
- Baselines trained (simple ML + DL); initial metrics logged.
- CNN experiment running (first results); NLP component scaffolded/prototyped.
- RL agent stubbed with reward design; early learning curves (even if noisy).
- Draft Model Card sections started; Ethics statement updated.
- **Deliver:** 1–2 page **Checkpoint Report** in docs/ + updated repo.

Week 3 — Final Submission & Defense (Due end of Week 3)

- Code **frozen** and reproducible (make repro or run.sh).
- Results tables/plots; ablations (≥2); error/slice analysis.
- **Final Report (4–6 pages)** conference-style PDF.
- **Slide Deck (8–10 slides)** and **Live Demo** (or 3–5 min recorded demo).
- **Defense**: 8–10 minutes presentation + 8–10 minutes Q&A.

6) GitHub Repository (Required Structure)

```
project-root/
  README.md                     # project overview, quick start, results
highlights, team
  LICENSE
  requirements.txt | environment.yml
  makefile | run.sh             # one-command reproduce
  data/
    README.md                   # how to obtain data (no raw PII in repo)
    get_data.py | download.sh
  src/
    data_pipeline.py
    models/                     # CNN/NLP architectures
    train.py
```

```
    eval.py
    rl_agent.py
    utils/
  notebooks/
    01_eda.ipynb
  experiments/
    configs/
    logs/                        # training logs, curves
    results/                     # tables, plots, figures
  docs/
    proposal.pdf
    checkpoint.pdf
    final_report.pdf
    slides.pdf
    model_card.md
    ethics_statement.md
```

For Automation it is recommended:

- Add a GitHub Action to run lint/tests (optional).
- Save seeds/configs; export training/eval metrics to CSV/JSON.
- Store figures (PR/ROC curves, learning curves) in experiments/results/.

7) Documentation Requirements

In your **Final Project Report**, clearly explain:

- **Problem** & motivation; success metrics and constraints.
- **Dataset** used (source, license, collection notes), preprocessing, and **splits**.
- **Training** details: hyperparameters, schedules, compute used, reproducibility steps.
- **Modeling**: architecture(s), why chosen, CNN/NLP/RL components, and how they connect.
- **Evaluation**: metrics, baselines, **ablations (≥2)**, error/slice analysis, calibration (if relevant).
- **Ethics & Policy**: risks, mitigations, intended use & limitations, privacy, fairness checks.
- **Model Card** summary with deployment guidance/disclaimers.
- **How to run**: one-command reproduction, environment setup, and demo instructions.

8) Evaluation Rubric (100 points)

Category	Pts	Criteria
Proposal & Problem Framing	10	Clear objective, measurable metrics, scoped MVP, related work
Data & Preparation	12	Governance, licensing/consent, preprocessing, leakage prevention
Core DL Modeling	18	Sound architecture, justified design, training quality
CNN Component	8	Correct implementation, analysis, and insights
NLP Component	8	Meaningful NLP piece (core or auxiliary) with evaluation

Category	Pts	Criteria
RL Component	8	Working agent, reward design, learning curves, integration
Experiments & Evaluation	16	Baselines, **≥2 ablations**, robust metrics, error/slice analysis
Ethics & Model Card	10	Concrete risks & mitigations, limitations, responsible guidance
Reproducibility & Engineering	6	One-command repro, environment capture, organized repo
Presentation & Defense	4	Clear slides/demo, confident Q&A

Individual contribution: Commit history, issues, PRs, and peer assessments may adjust individual scores by ±10%.

9) Minimum Technical Expectations

- **At least one model from scratch** (small) to demonstrate fundamentals, even if you also fine-tune a pretrained model.
- **Metrics**:
  - Classification: Accuracy + macro-F1; confusion matrix; (optional) calibration.
  - Detection/Segmentation: mAP/mIoU; PR curves.
  - NLP: Macro-F1 (classification), ROUGE/BLEU (generation), retrieval metrics where relevant.
  - RL: Learning curves (reward vs. steps/episodes), success rate, variance over ≥3 seeds (if time permits).
- **Runtime**: Full pipeline (train+eval) must complete in **≤ 90 minutes** in your environment configuration.

10) Defense Format & Expectations

- **8–10 min presentation**: problem, data, methods (CNN/NLP/RL), results, ethics, and demo.
- **Demo**: reliable, short; recorded fallback acceptable (if live presentation is not possible)
- **Q&A (8–10 min)**: be ready to discuss design trade-offs, ablations, failure cases, and mitigations.
- Bring learning curves, PR/ROC, confusion, and examples of errors and how you addressed them.

11) Academic Integrity, Safety & Accessibility

- Cite all datasets, code, and pre-trained models. Mark external code clearly.
- No hidden data collection; remove/obfuscate any PII; obtain consent.
- Add disclaimers for sensitive domains (health, safety-critical).
- Ensure accessible slides/reports (contrast, alt text for essential figures, readable fonts).

12) Submission Checklist (End of Week 3)

- GitHub URL with **v1.0 release**

- README.md with quick start and results summary
- **Proposal (PDF)**, **Checkpoint (PDF)**, **Final Report (PDF)** in docs/
- **Slides (PDF/PPTX)** and **Demo video** (if not live)
- requirements.txt/environment.yml and one-command repro script
- model_card.md and ethics_statement.md
- Ablation results and error/slice analysis plots